Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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4 Direct Proof

Use the method of direct proof to prove the following statements.

2. If x is an odd integer, then x^3 is odd.

Proof. Suppose x is an odd integer. Thus $x = 2a + 1$ for some $a \in \mathbb{Z}$. Consequently $x^3 = (2a+1)^3 = 8a^3+12a^2+6a+1 = 2(4a^3+6a^2+3a)+1$. Thus $x^3 = 2b+1$, where b is the integer $4a^3 + 6a^2 + 3a$. Therefore x^3 is odd by definition of an odd number. □

4. Suppose $x, y \in \mathbb{Z}$. If x and y are odd, then xy is odd.

Proof. Suppose x and y are odd. Then $x = 2a + 1$ for some $a \in \mathbb{Z}$, and $y = 2b + 1$ for some $b \in \mathbb{Z}$. Consequently $xy = (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$. Thus $xy = 2c + 1$, where c is the integer $2ab + a + b$. Therefore xy is odd by definition of an odd number. □

6. Suppose $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

Proof. Suppose $a \mid b$ and $a \mid c$. Then $b = ak$ for some $k \in \mathbb{Z}$ and $c = ah$ for some $h \in \mathbb{Z}$. Consequently $b + c = ak + ah = a(k + h)$. Thus $b + c = am$, where m is the integer $k + h$. Therefore $a \mid (b + c)$ by definition of divisibility. □

8. Suppose a is an integer. If $5 \mid 2a$, then $5 \mid a$.

Proof. Suppose $5 \mid 2a$. Then $2a = 5k$ for some $k \in \mathbb{Z}$. Observe that $a = 6a - 5a = 3(2a) - 5a$. Substituting $2a = 5k$ gives $a = 3(5k) - 5a = 15k - 5a = 5(3k - a)$. Thus $a = 5n$, where n is the integer $3k - a$. Therefore $5 \mid a$ by definition of divisibility. □

10. Suppose a and b are integers. If $a \mid b$, then $a \mid (3b^3 - b^2 + 5b)$.

Proof. Suppose $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Consequently $3b^3 - b^2 + 5b = 3(ak)^3 - (ak)^2 + 5(ak) = a(3k(ak)^2 - ak^2 + 5k)$. Thus $3b^3 - b^2 + 5b = am$, where m is the integer $3k(ak)^2 - ak^2 + 5k$. Therefore $a \mid (3b^3 - b^2 + 5b)$ by definition of divisibility. $\square$

12. If $x \in \mathbb{R}$ and $0 < x < 4$, then $\frac{4}{x(4-x)} \geq 1$.

Proof. Suppose $0 < x < 4$. Then $x > 0$ and $4 - x > 0$. This means that $x(4-x) > 0$, so we can multiply both sides of the inequality by $x(4-x)$, then it is enough to show that $4 \geq x(4-x)$.

Now, $x(4-x) = 4x - x^2$, hence $4 - (4x - x^2) = x^2 - 4x + 4 = (x-2)^2$. Because the square of any real number is positive, $(x-2)^2 \geq 0$, therefore $4 - (4x - x^2) \geq 0$ which implies $4 \geq x(4-x)$. Because $x(4-x) > 0$, dividing both sides by $x(4-x)$ gives $\frac{4}{x(4-x)} \geq 1$. $\square$

14. If $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd. (Try cases.)

Proof. Suppose $n \in \mathbb{Z}$. Every integer is either even or odd, since dividing an integer by 2 leaves a remainder of either 0 or 1. The following two cases are therefore exhaustive.

Case 1. Suppose n is even. Then $n = 2a$ for some $a \in \mathbb{Z}$. Consequently $5n^2 + 3n + 7 = 5(2a)^2 + 3(2a) + 7 = 20a^2 + 6a + 7 = 2(10a^2 + 3a + 3) + 1$. Thus $5n^2 + 3n + 7 = 2m + 1$, where m is the integer $10a^2 + 3a + 3$. Hence, in this case, $5n^2 + 3n + 7$ is odd by definition of an odd number.

Case 2. Suppose n is odd. Then $n = 2b + 1$ for some $b \in \mathbb{Z}$. Consequently $5n^2 + 3n + 7 = 5(2b+1)^2 + 3(2b+1) + 7 = 5(4b^2 + 4b + 1) + 6b + 10 = 20b^2 + 26b + 15 = 2(10b^2 + 13b + 7) + 1$. Thus $5n^2 + 3n + 7 = 2m + 1$, where m is the integer $10b^2 + 13b + 7$. Hence, in this case, $5n^2 + 3n + 7$ is odd by definition of an odd number.

The two cases are exhaustive and $5n^2 + 3n + 7$ is odd in each, so $5n^2 + 3n + 7$ is odd for every $n \in \mathbb{Z}$. $\square$

16. If two integers have the same parity, then their sum is even. (Try Cases.)

Proof. Suppose n and m are integers and they have the same parity. Then either both are even or both are odd, by definition of parity. We must therefore consider both cases.

Case 1. Assume n and m are even. Then $n = 2a$ and $m = 2b$ for some $a, b \in \mathbb{Z}$. Hence, $n + m = 2a + 2b = 2(a + b)$. Because $n + m = 2c$ where c is the integer $a + b$, their sum is even by definition of an even number.

Case 2. Assume n and m are odd. Then $n = 2a + 1$ and $m = 2b + 1$ for some $a, b \in \mathbb{Z}$. Hence, $n + m = 2a + 2b + 2 = 2(a + b + 1)$. Because $n + m = 2c$ where c is the integer $a + b + 1$, their sum is even by definition of an even number.

The two cases are exhaustive and they show that if two integers have the same parity then their sum is even. $\square$

18. Suppose x and y are positive real numbers. If $x < y$, then $x^2 < y^2$.

Proof. Suppose x and y are positive real numbers with $x < y$. Subtracting y from both sides gives $x - y < 0$. Because $x > 0$ and $y > 0$, the sum $x + y$ is positive, so multiplying both sides of $x - y < 0$ by $x + y$ preserves the inequality and gives $(x - y)(x + y) < 0$. Expanding the left side we get $x^2 - y^2 < 0$. Adding y^2 to both sides gives $x^2 < y^2$. $\square$

20. If a is an integer and $a^2 \mid a$, then $a \in \{-1, 0, 1\}$.

Proof. Suppose a is an integer and $a^2 \mid a$. Then $a = a^2 k$ for some $k \in \mathbb{Z}$. Either $a = 0$ or $a \neq 0$, so the following two cases are exhaustive.

Case 1. Suppose $a = 0$. Then $a \in \{-1, 0, 1\}$, since 0 is an element of that set.

Case 2. Suppose $a \neq 0$. Dividing both sides of $a = a^2 k$ by a gives $1 = ak$. Thus $a \mid 1$ by definition of divisibility. Taking absolute values, $|a||k| = 1$, and since $|a|$ and $|k|$ are positive integers, this forces $|a| = 1$. Hence $a \in \{-1, 1\}$, and so $a \in \{-1, 0, 1\}$.

The two cases are exhaustive and $a \in \{-1, 0, 1\}$ in each. Therefore $a \in \{-1, 0, 1\}$. $\square$

22. If $n \in \mathbb{N}$, then $n^2 = 2\binom{n}{2} + \binom{n}{1}$. (You may need a separate case for $n = 1$)

Proof. Suppose $n \in \mathbb{N}$. Then $n = 1$ or $n > 1$. We consider both cases.

Case 1. Suppose $n = 1$. Then $2\binom{n}{2} + \binom{n}{1} = 2\binom{1}{2} + \binom{1}{1} = 0 + 1 = 1 = 1^2 = n^2$. We see the condition holds for $n = 1$.

Case 2. Suppose $n > 1$. Then $2\binom{n}{2} + \binom{n}{1} = 2\left(\frac{n!}{2!(n-2)!}\right) + \left(\frac{n!}{1!(n-1)!}\right) = \frac{2n!}{2!(n-2)!} + \frac{n!}{(n-1)!} = \frac{n!}{(n-2)!} + \frac{n!}{(n-1)!} = n(n-1) + n = n^2$. Thus the condition holds for $n > 1$.

The two cases are exhaustive and show that $n^2 = 2\binom{n}{2} + \binom{n}{1}$. □

24. If $n \in \mathbb{N}$ and $n \geq 2$, then the numbers $n!+2, n!+3, n!+4, n!+5, ..., n!+n$ are all composite. (Thus for any $n \geq 2$, one can find n consecutive composite numbers. This means there are arbitrarily large "gaps" between prime numbers.)

Proof. Suppose $n \in \mathbb{N}$ and $n \geq 2$. Now consider the numbers $n!+2$, $n!+3$, ..., $n!+n$. Now take any $k \in \mathbb{N}$ with $2 \leq k \leq n$. Because $n! = 1 \cdot 2 \cdot ... \cdot n$ and $2 \leq k \leq n$, the number k occurs as one of the factors of $n!$. Hence $n! = km$, where m is the product of the remaining factors. Each of those factors is a positive integer, so m is a positive integer, and therefore $m \geq 1$. Then $n! + k = km + k = k(m+1)$. Now $k \geq 2 > 1$, and $m + 1 \geq 2 > 1$. Thus $n! + k$ is a product of two natural numbers, each greater than 1, so $n! + k$ is composite by definition of a composite number.

Since k was an arbitrary natural number with $2 \leq k \leq n$, every number in the list $n! + 2, n! + 3, ..., n! + n$ is composite. □